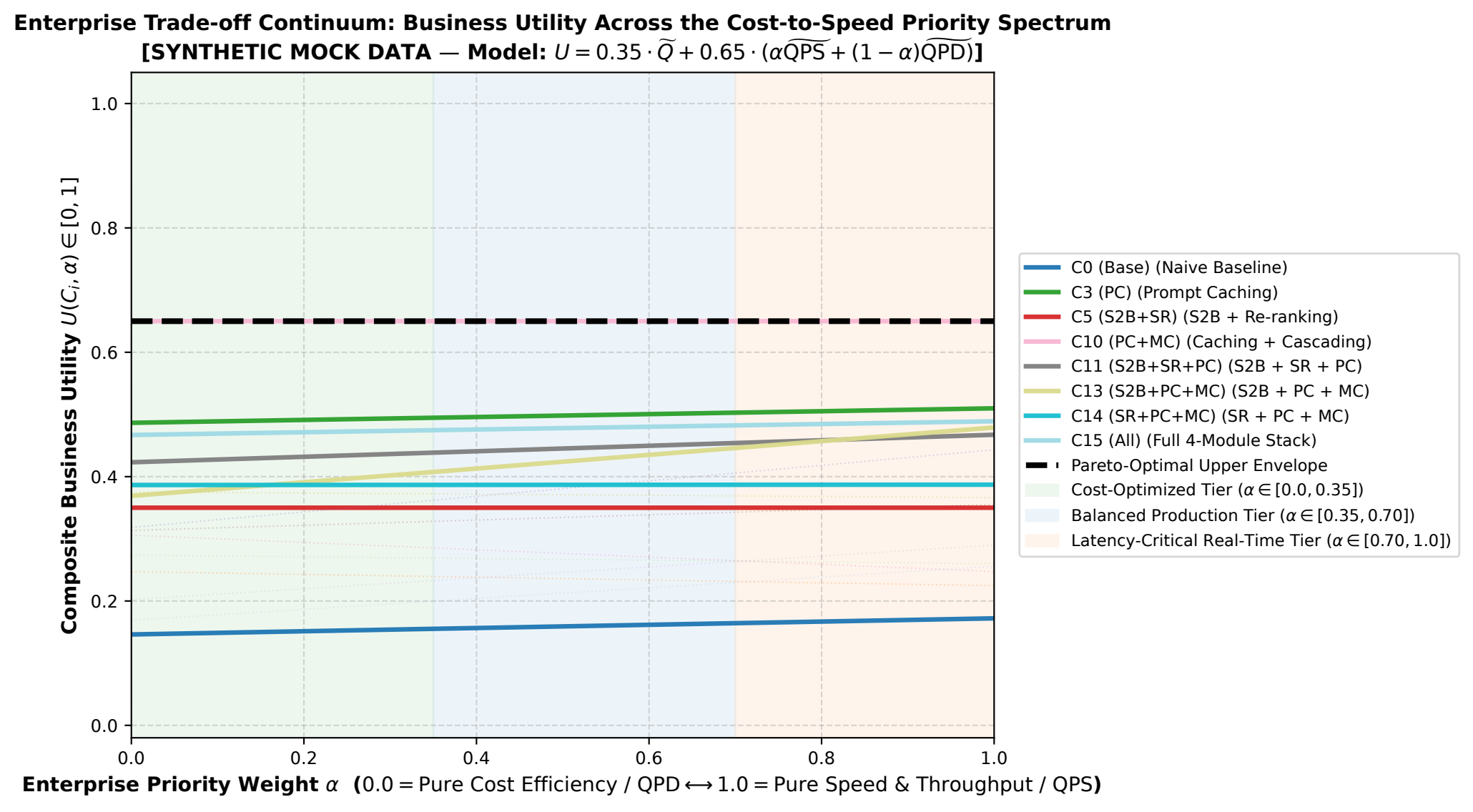
Enterprise Trade-off Continuum: Business Utility Across the Cost-to-Speed Priority Spectrum
[SYNTHETIC MOCK DATA — Model: $U = 0.35 \cdot \widetilde{Q} + 0.65 \cdot (\alpha\widetilde{\text{QPS}} + (1-\alpha)\widetilde{\text{QPD}})$]
Composite Business Utility $U(C_i, \alpha) \in [0, 1]$
0.0
0.2
0.4
0.6
0.8
1.0
Enterprise Priority Weight α (0.0 = Pure Cost Efficiency / QPD ⟷ 1.0 = Pure Speed & Throughput / QPS)
C0 (Base) (Naive Baseline)
C3 (PC) (Prompt Caching)
C5 (S2B+SR) (S2B + Re-ranking)
C10 (PC+MC) (Caching + Cascading)
C11 (S2B+SR+PC) (S2B + SR + PC)
C13 (S2B+PC+MC) (S2B + PC + MC)
C14 (SR+PC+MC) (SR + PC + MC)
C15 (All) (Full 4-Module Stack)
Pareto-Optimal Upper Envelope
Cost-Optimized Tier ($\alpha \in [0.0, 0.35]$)
Balanced Production Tier ($\alpha \in [0.35, 0.70]$)
Latency-Critical Real-Time Tier ($\alpha \in [0.70, 1.0]$)